Generative AI & Prompt Engineering for Software Development

MODULE 1: Foundations of Generative AI (Must-Know Basics)

1.1 Introduction to Generative AI

- What is Generative AI?
- Difference: AI vs ML vs Deep Learning vs Generative AI
- How Large Language Models (LLMs) work (high level)
- What AI **can** and **cannot** do

1.2 Role of Generative AI in Software Development

- Traditional SDLC vs AI-assisted SDLC
- Where AI fits in:
    - Planning
    - Coding
    - Testing
    - Documentation
    - Deployment
- Human-in-the-loop concept

MODULE 2: Prompt Engineering (Practical, Not Hype)

2.1 What Prompt Engineering Really Is

- Prompt engineering = clear requirement communication
- Why prompts fail
- AI hallucinations & how to avoid them

2.2 Prompt Structures That Work

- Context + Task + Constraints + Output format
- Examples:
    - Coding prompts
    - Debugging prompts

    - Documentation prompts

2.3 Iterative Prompting

- Refining prompts step-by-step
- Asking follow-up prompts
- Validating AI output

MODULE 3: AI for Requirement Analysis & Documentation

3.1 Requirement Understanding

- Convert business ideas into:
    - Functional requirements
    - Non-functional requirements
- User stories & acceptance criteria using AI

3.2 Technical Documentation with AI

- API documentation
- Architecture descriptions
- README files
- Release notes

3.3 Tools

- ChatGPT
- Notion AI
- Mintlify / Swimm AI

MODULE 4: AI-Assisted Coding (Core Development Skill)

4.1 Code Generation

- Writing functions, classes, modules
- CRUD APIs
- Database queries
- Frontend components

4.2 Supported Tech Stacks

- Java (Spring Boot)
- Python (Django / Flask)
- JavaScript (React / Node)
- SQL

4.3 Tools

- GitHub Copilot
- ChatGPT
- Amazon CodeWhisperer

MODULE 5: AI for Code Review, Refactoring & Optimization

5.1 Code Quality Improvements

- Naming conventions
- Removing duplicate code
- Clean architecture suggestions

5.2 Performance Optimization

- Algorithm improvements
- Query optimization
- Memory & time efficiency suggestions

5.3 Tools

- Sourcegraph Cody
- SonarQube (AI insights)
- Snyk Code

MODULE 6: Debugging & Error Resolution Using AI

6.1 Debugging Techniques

- Reading stack traces
- Log analysis
- Runtime vs compile-time errors

6.2 AI-Assisted Debugging

- Explaining errors using AI
- Fix suggestions
- Root cause analysis

6.3 Tools

- ChatGPT
- GitHub Copilot Chat
- Sentry AI

MODULE 7: Automated Testing with Generative AI

7.1 Importance of Testing

- Unit testing
- Integration testing
- Edge-case testing

7.2 AI-Generated Test Cases

- Writing test cases from code
- Mock data generation
- Coverage improvement

7.3 Tools

- Diffblue Cover (Java)
- Copilot test generation
- Testim

MODULE 8: UI/UX Development with AI

8.1 AI in Frontend Development

- Converting ideas → UI
- Design → code
- Rapid prototyping

8.2 Tools

- Vercel v0
- Figma AI
- Uizard

MODULE 9: DevOps, CI/CD & Deployment Using AI

9.1 Deployment Basics

- CI/CD concepts
- Docker & cloud basics (high level)

9.2 AI-Assisted DevOps

- Writing CI/CD pipelines
- Dockerfile generation
- Cloud deployment scripts

9.3 Tools

- GitHub Copilot CLI
- ChatGPT
- Harness AI

MODULE 10: Security, Ethics & Limitations (Very Important)

10.1 AI Risks

- Security vulnerabilities
- Outdated code
- Data leakage risks

- Licensing issues

10.2 Responsible AI Usage

- Code validation
- Security reviews
- Ethical considerations

MODULE 11: End-to-End AI-Assisted Project

Example Project

Build a Full-Stack Application using Generative AI

- Requirement documentation using AI
- Backend API development
- Frontend UI
- Testing
- Documentation
- Deployment plan